

Problem Approach & Solution Design

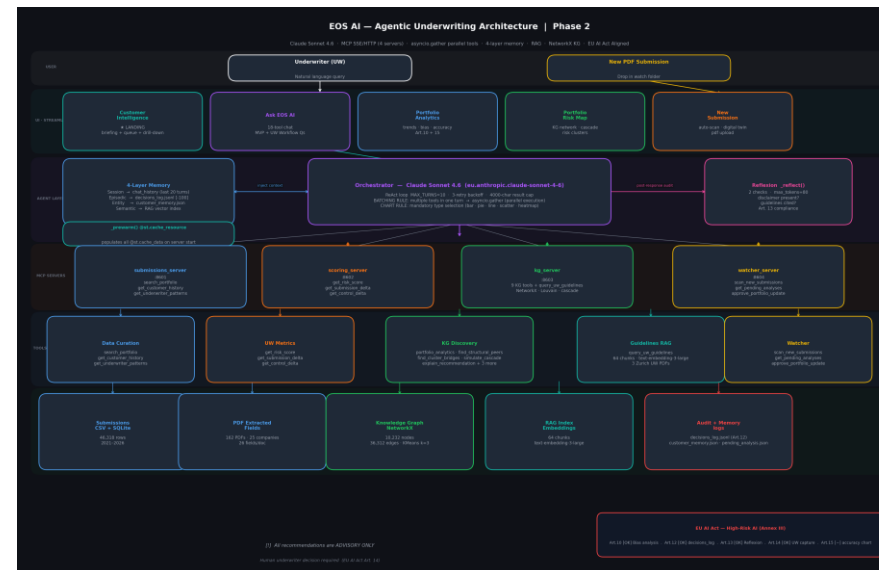
From 80 minutes to 3 minutes — EOS AI for Repeat Submissions

Drawback

- Zurich Specialties treats repeat customers as new business every year
- UW spends in avg 55–80 min per submission doing manual research
- 46,318 submissions (2021–2026) contain the answer — unused

Advantage:

- EOS AI: 18-tool ReAct agent (Claude Sonnet 4.6)
- Knowledge Graph: 10,232 nodes · 36,312 edges
- 4 MCP servers · asyncio.gather parallel execution
- RAG on 3 Zurich UW guideline PDFs
- Event-driven watcher: auto-analyses new PDFs in 30s
- EU AI Act Art.10/12/13/14/15 compliant from day 1



Results & Key Insights

Validated results on 9,078 repeat customers

Main Metrics :

- 🕒 55–80 min → < 3 min per repeat submission
- 📊 34% of book are repeat customers (9,078 identified)
- 🎯 Bind rate proof: FAST_TRACK 68% · STANDARD_UW 41% · FRESH_UW 12%

Additional Key insights:

- Knowledge Graph reveals hidden cascade risks (broker failure → portfolio contagion)
- Digital twin matching: new submissions matched to structural peers in real time
- Chain of thought visible to UW — no black box
- Full audit trail in decisions_log.jsonl (EU AI Act Art. 12)

Next Steps & Recommendations

Production roadmap — three phases, clear governance

Horizon 1 — Production Ready (0–3 months)

- Docker/K8s for all 4 MCP servers · PostgreSQL · Azure AD SSO
- CI/CD pipeline: lint → tests → staging → production
- Observability: structured logging · token cost tracking · latency dashboards
- Azure Key Vault for secrets · role-based access (UW / Team Lead / Admin)

Horizon 2 — Scale & Reliability (3–6 months)

- React + FastAPI frontend · WebSocket streaming · Agent Gateway with rate limiting
- Kafka event-driven ingestion (replace 30s file watcher)
- Neo4j persistent KG with timestamped edges · `snapshot\_at(date)` queries
- Azure AI Search managed vector store · automated RAG re-indexing on guideline updates
- Cost at scale: ****~\$375/month**** LLM · **~\$200–500/month** infrastructure

Horizon 3 — Intelligent System (6–12 months)

- ****Multi-agent:**** Orchestrator + Data Agent + Risk Agent + KG Agent + Governance Agent
- Feedback loop from aggregated UW decisions (GDPR Art. 22 compliant)
- External signals: threat intelligence feeds · financial news → KG enrichment
- EU AI Act Art. 43 conformity assessment (third-party audit · ~€15–30K · 4–8 weeks)
- Monthly drift detection · DPIA · RACI documented

*"We built EU AI Act governance from day one. No surprises for legal."**